■ EPL SQL Challenge Questions

English Premier League · 1993–2017 · MySQL

Dataset: The main table is `EPL` with key columns: `HomeTeam`, `AwayTeam`, `FTHG` (home goals), `FTAG` (away goals), `FTR` (result: H/A/D), `Season`, `MatchDate`.

■ Warm Up — SELECT / WHERE / ORDER BY

1. Show all matches where the home team scored 6 or more goals (`FTHG >= 6`).

■ Display: MatchDate, HomeTeam, AwayTeam, FTHG, FTAG.

2. List all distinct teams that have ever played as a home team, sorted alphabetically.

3. Find all matches played in the `'2003-2004'` season where the result was a draw (`FTR = 'D'`).

■ Aggregation — GROUP BY / HAVING

4. How many total matches were played each season? Show `Season` and match count, ordered by season.

5. Which home teams have scored more than 100 home goals across all seasons? Use `SUM(FTHG)`.

6. Find teams where their average away goals scored (`AVG(FTAG)`) is greater than 1.5 — but only include teams with at least 50 away games.

■ CASE Statements

7. Add a `result_label` column to every match that says `'Home Win'`, `'Away Win'`, or `'Draw'` based on `FTR`.

8. Categorize each match by goal excitement based on total goals (`FTHG + FTAG`):

'Thriller' → 5+ goals | 'Good Game' → 3–4 goals | 'Boring' → 2 or fewer goals

■ String Functions

9. The `Season` column looks like `'2003-2004'`. Use `SUBSTRING()` or `LEFT()` to extract just the starting year (e.g. `'2003'`).

10. Find all matches where the home team name contains the word `'United'`.

■ Joins

First, create a small lookup table by running this SQL:

```
CREATE TABLE Teams (
  TeamName VARCHAR(50),
```

```
  City      VARCHAR(50),
  Founded   INT
);

INSERT INTO Teams VALUES
  ('Arsenal',           'London',     1886),
  ('Chelsea',           'London',     1905),
  ('Manchester United', 'Manchester', 1878),
  ('Manchester City',   'Manchester', 1880),
  ('Liverpool',         'Liverpool',  1892),
  ('Tottenham',         'London',     1882);
```

11. Join `EPL` to `Teams` on `HomeTeam = TeamName` to show all home matches for London-based clubs only.

Subqueries

12. Find all matches where `FTHG` was greater than the **average home goals** across the entire dataset.

13. Find all teams that have **never** won a single away game. Use a subquery with `NOT IN`.

Hint: They won't appear in AwayTeam where FTR = 'A'

UNION

14. Create a combined list of the **top 5 home scorers** and **top 5 away scorers** (by total goals). Add a column called `type` that says `'Home'` or `'Away'` to label each group.

Window Functions

15. Rank all teams by total home wins **per season** using `RANK()`.

Partition by Season, order by win count descending — so you can see the best home team each year.

16. For Arsenal's home matches (ordered by `MatchDate`), show a **running total** of their home goals using `SUM(FTHG) OVER (ORDER BY MatchDate)`.

17. Use `LAG()` to compare each home team's goals in a match vs. their **previous home match** — so you can see if they scored more or fewer than last time.

Tips

- Work through the questions **in order** — later ones build on earlier intuitions.
- For questions 16–18, try writing the `PARTITION BY` / `ORDER BY` clause first and see what comes out.
- Q12 (self-join) and Q15 (UNION) are the trickiest — save those for last.
- If you get stuck, try breaking the query into smaller pieces and testing each part separately.